

OSSP:

Commands:

uname : Displays basic operating system information.

```
vedadripinninti@vedadris-MacBook-Air ~ % nano Task.c
vedadripinninti@vedadris-MacBook-Air ~ % gcc Task.c -o Task
vedadripinninti@vedadris-MacBook-Air ~ % ./Task
Enter a Linux command: ls

Parent Process
Parent PID: 10360
Child PID: 10361

Child Process
Child PID: 10361
Parent PID: 10360
<!doctype html>.html          Home page.html          PiDs
2ebc9504-640c-4e44-9851-d67271161b01.JPG  IMG_8048.jpg            PIDs.c
airline                        IMG_8049.heic            process
Applications                  IMG_8050.JPG            process.c
Desktop                       Library                 project1.1
Documents                     Movies                  Public
Downloads                     Music                   Task
file                           Passbook.jpg            Task.c
file.c                         Pictures                 Task.c.save
Child process completed.
```

lscpu : Displays CPU and processor details.

```
/Users/vedadripinninti/PiDs ; exit;
vedadripinninti@vedadris-MacBook-Air ~ % /Users/vedadripinninti/PiDs ; exit;
Enter a Linux command: ps

Parent Process
Parent PID : 10497
Child PID  : 10498

Child Process
Child PID  : 10498
Parent PID : 10497
  PID TTY          TIME CMD
 10349 ttys000      0:00.04 -zsh
  10492 ttys001      0:00.01 -zsh
  10497 ttys001      0:00.00 /Users/vedadripinninti/PiDs
Child process completed.

Saving session...
...copying shared history...
...saving history...truncating history files...
...completed.

[Process completed]
```

ps : Displays the currently running processes.

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system manages the CPU by dividing its processing time among multiple applications.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It processes both Chrome and music player.

2. Memory (RAM) -> Memory Management

The operating system assigns a separate portion of RAM to every running application.

Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system stores and organizes files on the storage device.

Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.